Functions: Modulus & Inverse

ANSWER KEY

Section 1: Modulus Equations & Linear Graphs

1 Solve the modulus equation $|3x - 5| = 2x + 1$ for all real solutions.

a. $x = 6$ **and** $x = 0.8$ ✓

b. $x = 6$ only

c. $x = 4$ and $x = -2$

d. $x = 0.8$ only

Case 1: $3x - 5 = 2x + 1 \Rightarrow x = 6$. Check: $|13| = 13$ (valid). Case 2: $3x - 5 = -(2x + 1) \Rightarrow 5x = 4 \Rightarrow x = 4/5 = 0.8$. Check: $|-2.6| = 2.6 = 2(0.8) + 1$ (valid). Both are solutions.

2 Solve the modulus equation $|2x - 7| = |x + 2|$.

a. $x = 9$ **and** $x = 5/3$ ✓

b. $x = 9$ only

c. $x = 5$ and $x = -3$

d. $x = 3$ and $x = -1$

Square both sides: $(2x - 7)^2 = (x + 2)^2 \Rightarrow (2x - 7 - x - 2)(2x - 7 + x + 2) = 0 \Rightarrow (x - 9)(3x - 5) = 0 \Rightarrow x = 9$ or $x = 5/3$.

3 Find the coordinates of the vertex of the graph $y = |2x - 6| + 4$.

a. $(3, 4)$ ✓

b. $(-3, 4)$

c. $(3, -4)$

d. $(6, 4)$

The vertex occurs where the argument of the modulus equals zero: $2x - 6 = 0 \Rightarrow x = 3$. At $x = 3$, $y = 0 + 4 = 4$. Vertex is $(3, 4)$.

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Section 2: Modulus Inequalities

4 Solve the inequality $|2x - 5| < 7$.

a. $-1 < x < 6$ ✓

b. $x < -1$ or $x > 6$

c. $-6 < x < 1$

d. $1 < x < 6$

$-7 < 2x - 5 < 7 \Rightarrow -2 < 2x < 12 \Rightarrow -1 < x < 6$.

5 Solve the inequality $|3x + 1| \geq |x - 4|$.

a. $x \leq -5/2$ **or** $x \geq 3/4$ ✓

b. $-5/2 \leq x \leq 3/4$

c. $x \leq -1$ or $x \geq 4$

d. $x \geq 3/4$

Square both sides: $(3x + 1)^2 \geq (x - 4)^2 \Rightarrow (3x + 1)^2 - (x - 4)^2 \geq 0 \Rightarrow (4x - 3)(2x + 5) \geq 0$. Roots at $3/4$ and $-5/2$. Product is non-negative outside roots: $x \leq -5/2$ or $x \geq 3/4$.

6 Find the set of values of x satisfying $|x - 2| > 2x - 1$.

a. $x < 1$ ✓

b. $x > 1$

c. $-1 < x < 1$

d. $x < 3$

For $x \geq 2$: $x - 2 > 2x - 1 \Rightarrow x < -1$ (no solution in $[2, \infty)$). For $x < 2$:
$-(x - 2) > 2x - 1 \Rightarrow -x + 2 > 2x - 1 \Rightarrow 3x < 3 \Rightarrow x < 1$.

Section 3: Composite and Inverse Functions in A2

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7 The function $f(x) = 2 + \ln(x - 3)$ is defined for $x > 3$. Find the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = 3 + e^{x-2}$, **domain** $x \in \mathbb{R}$ ✓

b. $f^{-1}(x) = 3 - e^{x-2}$, domain $x > 0$

c. $f^{-1}(x) = 2 + e^{x-3}$, domain $x \in \mathbb{R}$

d. $f^{-1}(x) = \ln(x - 2) + 3$, domain $x > 2$

Let $y = 2 + \ln(x - 3) \Rightarrow \ln(x - 3) = y - 2 \Rightarrow x - 3 = e^{y-2} \Rightarrow x = 3 + e^{y-2}$. The range of f is $\mathbb{R}$, so domain of f^{-1} is $\mathbb{R}$.

8 Given $f(x) = e^{2x}$ for $x \in \mathbb{R}$ and $g(x) = \dfrac{1}{x}$ for $x > 0$, solve the equation $fg(x) = e^4$.

a. $x = 1/2$ ✓

b. $x = 2$

c. $x = 1/4$

d. $x = 4$

$fg(x) = f(1/x) = e^{2/x} = e^4 \Rightarrow 2/x = 4 \Rightarrow x = 2/4 = 1/2$.

9 A function $h(x) = x^2 - 4x + 1$ is defined on the domain $x \leq k$. Find the greatest value of k for which h is one-to-one.

a. $k = 2$ ✓

b. $k = 4$

c. $k = 0$

d. $k = 1$

The parabola has vertex at $x = -(-4)/(2) = 2$. For $x \leq k$ to be strictly decreasing (one-to-one), the domain must not extend past the vertex, so greatest $k = 2$.

Section 4: Transformations & Analytical Modeling

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10 Describe the transformation mapping the graph of $y = \ln(x)$ onto the graph of $y = \ln(3x - 6)$.

a. **Horizontal stretch scale factor $1/3$ followed by translation $[2, 0]$** ✓

b. Translation $[6, 0]$ followed by horizontal stretch scale factor 3

c. Horizontal stretch scale factor 3 followed by translation $[-2, 0]$

d. Vertical stretch scale factor 3 followed by translation $[2, 0]$

Write as $y = \ln[3(x - 2)]$. The factor 3 represents a horizontal stretch by scale factor $1/3$, and $(x - 2)$ represents a translation of 2 units in the positive x-direction ($[2, 0]$).

11 Find the number of real roots of the equation $|2x - 3| = x^2 - 2$.

a. **2 roots** ✓

b. 4 roots

c. 0 roots

d. 1 root

For $x \geq 1.5$: $2x - 3 = x^2 - 2 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$ (not in $x \geq 1.5$). For $x < 1.5$: $-2x + 3 = x^2 - 2 \Rightarrow x^2 + 2x - 5 = 0 \Rightarrow x = \frac{-2 \pm \sqrt{24}}{2} = -1 \pm \sqrt{6}$. $-1 + \sqrt{6} \approx 1.45 < 1.5$ (valid), and $-1 - \sqrt{6} \approx -3.45 < 1.5$ (valid). Exactly 2 roots.

12 The graph of $y = f(x)$ passes through $(2, 5)$. What are the coordinates of the corresponding point on $y = 3 - 2f(x + 1)$?

a. $(1, -7)$ ✓

b. $(3, -7)$

c. $(1, 13)$

d. $(3, 7)$

New $x = 2 - 1 = 1$. New $y = 3 - 2(5) = 3 - 10 = -7$. Point is $(1, -7)$.

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